

TULLOCH BRIDGE, United Kingdom

WMO: 030470

Lat: 56.8669N

Lon: 4.7081W

Elev: 249

StdP: 98.37

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.5	(c) -5.5	(d) -9.4	(e) 1.7	(f) -6.8	(g) -6.8	(h) 2.2	(i) -2.5	(j) 13.3	(k) 6.1	(l) 11.8	(m) 5.9	(n) 0.4	(o) 0	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.6	(c) 22.1	(d) 15.7	(e) 20.0	(f) 14.6	(g) 18.3	(h) 14.0	(i) 16.7	(j) 20.4	(k) 15.7	(l) 18.8	(m) 14.8	(n) 17.3	(o) 3.8	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.3	11.2	17.9	14.4	10.5	17.0	13.5	9.9	16.2	47.9	20.6	44.7	18.9	42.2	17.4	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.6	(c) 7.6	(e) -11.1	(f) 25.3	(g) 2.8	(h) 1.9	(i) -13.1	(j) 26.6	(k) -14.7	(l) 27.7	(m) -16.3	(n) 28.7	(o) -18.4	(p) 30.1		
(a) 10.0	(b) 8.6	(c) 7.6	(e) -11.4	(f) 18.9	(g) 3.3	(h) 1.9	(i) -13.8	(j) 20.3	(k) -15.7	(l) 21.4	(m) -17.5	(n) 22.5	(o) -19.9	(p) 23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	2.8	2.9	4.0	6.3	9.0	11.5	13.2	12.6	11.0	8.1	5.2	2.8
	DBStd	4.87	3.35	3.32	3.04	2.89	3.10	2.60	2.32	2.30	2.54	2.97	3.32	4.24
	HDD10.0	1254	222	198	187	115	56	12	1	4	18	73	144	224
	HDD18.3	3963	481	431	445	361	288	206	161	178	219	318	393	482
	CDD10.0	335	0	0	1	4	26	57	99	84	49	13	2	1
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0
	CDH23.3	30	0	0	0	0	3	12	10	5	0	0	0	0
	CDH26.7	2	0	0	0	0	0	2	0	0	0	0	0	0
Wind	WSAvg	3.2	3.8	3.8	3.7	3.2	3.1	2.9	2.7	2.6	2.9	3.3	3.2	3.4
Precipitation	PrecAvg	1651	219	158	143	94	89	78	86	102	122	172	180	207
	PrecMax	2175	446	352	372	158	214	136	156	223	242	306	345	523
	PrecMin	1100	64	32	19	39	21	28	37	16	29	51	72	20
	PrecStd	282	93	96	78	32	49	29	31	46	56	76	69	124
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	11.5	13.7	18.3	23.1	25.0	24.7	23.3	20.1	15.7	13.3	12.0
		MCWB	9.2	9.4	8.5	11.5	14.9	16.7	17.7	18.3	15.1	12.8	11.4	10.1
	2%	DB	9.7	10.1	10.8	15.0	20.2	20.7	22.1	19.6	17.9	14.4	11.8	10.8
		MCWB	8.2	8.3	8.5	9.7	12.9	14.7	16.4	15.4	14.4	12.2	10.1	9.2
	5%	DB	8.6	9.0	9.6	12.8	17.5	18.4	19.8	18.0	16.4	13.3	10.9	9.7
		MCWB	7.2	7.4	7.5	8.9	11.8	13.6	15.2	14.5	13.6	11.3	9.3	8.2
	10%	DB	7.5	7.7	8.5	11.1	14.9	16.3	17.8	16.7	15.1	12.4	9.9	8.4
		MCWB	6.1	6.1	6.7	8.1	10.5	12.4	14.4	13.8	12.7	10.5	8.4	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.0	10.1	11.9	15.9	17.2	18.6	18.5	16.6	13.5	12.0	10.5
		MCDB	10.7	11.2	12.1	15.6	21.8	22.9	23.1	21.6	18.6	14.9	13.0	11.7
	2%	WB	8.5	8.6	8.7	10.6	13.8	15.6	17.0	16.1	15.1	12.6	10.6	9.5
		MCDB	9.6	9.9	10.4	13.9	18.6	20.1	20.8	18.6	17.0	14.0	11.6	10.7
	5%	WB	7.4	7.6	7.8	9.5	12.3	14.1	15.9	15.2	14.1	11.8	9.5	8.4
		MCDB	8.5	8.9	9.4	12.3	16.2	17.6	19.0	17.3	15.9	13.1	10.6	9.6
	10%	WB	6.3	6.4	6.9	8.5	11.0	13.0	14.8	14.2	13.1	10.8	8.5	7.1
		MCDB	7.5	7.7	8.5	10.8	14.4	15.6	17.3	16.3	14.9	12.1	9.8	8.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	6.3	7.6	8.7	7.8	7.6	7.3	6.9	5.9	5.3	5.5
		MCDBR	5.3	5.5	7.7	10.9	14.2	12.3	11.9	9.4	8.7	6.4	5.7	6.0
		MCWBR	4.8	4.6	5.6	6.8	8.0	7.0	7.0	5.9	5.9	5.0	5.0	5.5
	5% WB	MCDBR	5.3	5.1	7.2	8.9	11.8	10.7	10.0	7.9	7.7	5.6	5.9	6.0
		MCWBR	5.0	4.6	5.6	5.9	7.0	6.5	6.2	5.5	5.9	5.0	5.4	5.6
Clear-Sky Solar Irradiance	taub	0.321	0.336	0.361	0.380	0.384	0.378	0.379	0.362	0.352	0.337	0.315	0.307	
	taud	2.157	2.248	2.247	2.254	2.296	2.350	2.358	2.412	2.437	2.390	2.254	2.120	
	Ebn at Noon	526	702	786	832	852	863	853	845	798	702	541	429	
	Edh at Noon	57	78	101	116	119	114	111	99	84	68	51	45	
All-Sky Solar Radiation	RadAvg	0.40	1.02	2.04	3.41	4.51	4.39	4.17	3.31	2.29	1.22	0.53	0.30	
	RadStd	0.06	0.14	0.18	0.27	0.40	0.47	0.43	0.27	0.18	0.16	0.05	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-1.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page